

Glassworks (glass)

Giorgio and Edoardo are planning to go on a tour of Venice, and they are particularly interested in Murano's famous colored Venetian glass.



Figure 1: The city of Murano.

There are N shops which sell glasswares and they are numbered from 0 to $N - 1$, the i -th shop sells only pieces of color C_i . Giorgio and Edoardo are interested in visiting some shop and buying some glasswares, they will be happy if they have bought glasswares of **at least 3 different colors**.

The fastest (and the only one) way to travel in Murano is by *gondola*, in particular there are M *gondolas*, each *gondola* can carry Giorgio and Edoardo from shop A_i to B_i and viceversa. However, *gondolas* are not free and each trip requires a ticket (each ticket is valid for one trip only, so taking the same *gondola* twice requires two tickets).

Giorgio and Edoardo can start and finish the tour in any shop, help them find the minimum number of tickets required to buy at least 3 glasswares of different colors, or tell that they can't.

Among the attachments of this task you may find a template file `glass.*` with a sample incomplete implementation.

Input

The first line contains the integers N and M , the number of shops and the number of available *gondolas*.

The second line contains N integers, C_i for $i = 0 \dots N - 1$.

The following M lines contain 2 integers A_i and B_i each, meaning that a *gondola* can travel between shops A_i and B_i .

Output

You need to write a single line with an integer: the minimum number of tickets Giorgio and Edoardo need or -1 if they can't visit shops with glasswares of 3 different colors.

Constraints

- $2 \leq N \leq 500\,000$.
- $1 \leq M \leq 500\,000$.
- $0 \leq C_i \leq N - 1$ for $i = 0 \dots N - 1$.
- $0 \leq A_i, B_i \leq N - 1$, $A_i \neq B_i$ for $i = 0 \dots M - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (20 points) $N, M \leq 500$.

- **Subtask 3** (20 points) $N, M \leq 1000$.

- **Subtask 4** (20 points) For each $i = 0 \dots N - 1$, there are at most two values of j for which $A_j = i$ or $B_j = i$.

- **Subtask 5** (40 points) No additional limitations.


Examples

input	output
6 6 2 0 1 1 2 1 0 2 1 3 1 5 2 4 2 5 3 5	3
5 3 0 0 1 2 3 0 1 2 1 3 4	-1

Explanation

In the **first sample case** it is possible to start the tour in shop 4, go to shop 2, then go to shop 5 and end the tour in shop 1.

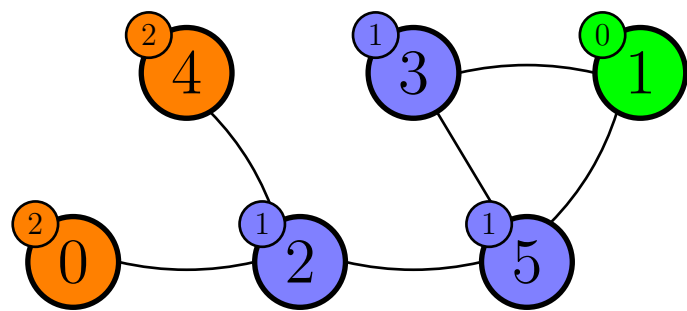


Figure 2: First sample case.

In the **second sample case** it isn't possible to plan a tour that passes by shops which have glass of 3 different colors.

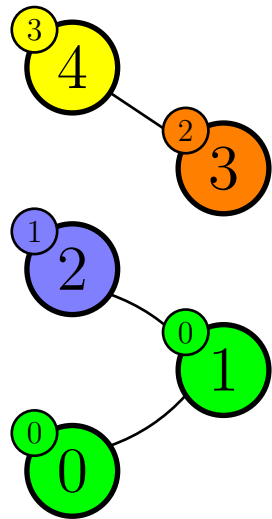


Figure 3: Second sample case.